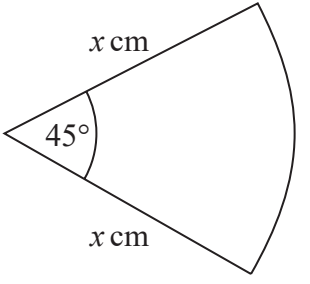
DO NOT WRITE IN THIS AREA

11 The diagram shows a sector of a circle of radius x cm.



The area of the sector is $72\pi\,\text{cm}^2$

Work out the value of x.

$\frac{45}{360} \times \pi x^2 = 72\pi$

1 mark
Sector area = angle/360 × π × radius squared.

$x^2 = 72 \times 8 = 576$, so $x = 24$ cm

1 mark
Take the positive square root for a length.

$x =$ 24

(Total for Question 11 is 2 marks)